

Reference Material for $\delta(^{13}\text{C})$

REFERENCE SHEET

IAEA-CH-6

NIST name: RM 8542

Sucrose, granular

Reference value of $\delta(^{13}\text{C})_{\text{VPDB-LSVEC}}$ of IAEA-CH-6 expressed in per mille (‰) on the VPDB-LSVEC scale and its associated combined standard uncertainty ($k=1$).

Parameter	Reference value	Combined standard uncertainty with $k=1$
$\delta(^{13}\text{C})^{(a)}_{\text{VPDB-LSVEC}} (\text{‰})$	-10.45 ^(b)	0.03 ^(b)

^(a) In this work, the expression of the quantity carbon isotope delta follows Skrzypek et al.[1] Alternative expressions for the quantity include $\delta(^{13}\text{C}/^{12}\text{C})$ [1] and more commonly $\delta^{13}\text{C}$ following published guidelines [2].

^(b) The reference value is calculated from four data sets, each being obtained by a different laboratory [3]. The uncertainty is expressed as combined uncertainty using a coverage factor $k = 1$ [3].

Origin and preparation of the material

IAEA-CH-6 (its former name was Sucrose ANU) was supplied to the IAEA by H. Polach, Australian National University, Canberra [4].

It was originally prepared by H.A. Polach, by selecting a batch of 2 tons of cane sucrose, grown between September 1965 and June 1971, which was sealed in drums (20kg each) to serve as a secondary ^{14}C standard [5].

Homogeneity of the material

A homogeneity test was performed in 2024 in IAEA using a sample size of 0.1 mg. The homogeneity test results and the data from inter-laboratory comparisons [3, 6] suggest that the contribution of heterogeneity to the combined uncertainty of the reference value is not significant.

Characterization study

The characterisation for $\delta(^{13}\text{C})$ value started in 1987 [4]. The current characterisation was performed in 2004 by a group of 4 experienced laboratories together with assessment and re-calibration of most of the other organic carbon stable isotope reference materials available at IAEA and published in 2006 [3].

Stability of the material

Since its release in 1983, no instances of instability have been reported for IAEA-CH-6 [6]. The value has been confirmed in 2023 [6] and at IAEA in 2024.

Assignment of values

The $\delta(^{13}\text{C})$ value has been determined using the Bayesian method and the uncertainty combines type A and B uncertainties. The details concerning the reported results may be found in references [3].

The $\delta(^{13}\text{C})$ value is the value accepted by the Commission on Isotopic Abundances and Atomic Weights of the International Union of Pure and Applied Chemistry (IUPAC) in 2014 [7].

Statement on metrological traceability and uncertainty of assigned values

The assigned value for $\delta(^{13}\text{C})$ is traceable to the VPDB-LSVEC scale normalised by assigning consensus values of -46.6‰ to LSVEC and $+1.95\text{‰}$ to NBS 19 [3].

For $\delta(^{13}\text{C})$, currently two scales exist and are in use: VPDB (Vienna Pee Dee belemnite) and VPDB-LSVEC (lithium carbonate prepared by H.J. Svec), as acknowledged by experts during the 12th IAEA meeting held in Vienna, 22-26 January 2024 [8]. $\delta(^{13}\text{C})$ values reported on the VPDB and VPDB-LSVEC scale can differ, with maximum reported differences (VPDB minus VPDB-LSVEC) of $+0.3\text{‰}$ for materials with carbon isotope delta values around -45‰ and negligible difference (below current analytical combined uncertainty) at carbon isotope delta values close to zero.

End users should state the values of reference materials used to normalize isotope data following the minimum requirements established by IUPAC [1] and whether they are publishing $\delta(^{13}\text{C})$ values on the VPDB scale or the VPDB-LSVEC scale. Users should

ensure that in one calibration they only use values for reference materials assigned on one scale.

Absolute isotopic abundances

No absolute isotope abundance ratio determination was performed on this material.

Intended use

IAEA-CH-6 is a secondary reference material. Its intended use is as calibrant when $\delta(^{13}\text{C})$ is determined in organic materials using IRMS (Isotope Ratio Mass Spectrometry) coupled with an EA (Elemental Analyser), for obtaining traceability of the data to the VPDB-LSVEC $\delta(^{13}\text{C})$ scale.

Instructions for use, handling and storage

IAEA-CH-6 is issued in units of 0.5 g.

It is recommended that it is stored in a dry, cool place protected from light in the tightly closed container in which it was supplied to the user.

Upon opening and taking a portion for the analysis, the container must be rapidly and tightly closed.

The recommended minimum sample size to be used is 0.1 mg.

Limit of distribution

Only one unit per year per laboratory may be ordered. This strategy should ensure that material is available for international use for as long as possible.

Issue and expiry date

The issue date of IAEA-CH-6 was 1983/1/9. The current characterization for $\delta(^{13}\text{C})$ has been published in 2006 [3].

The issue date of this reference sheet is provided in the footer of this document.

The assigned $\delta(^{13}\text{C})$ value was confirmed in 2023 [6] and in 2024. Based on this, the expiry date is 11/12/2034, provided the original packages are handled and stored in accordance with the instructions given in this reference sheet (see “Instructions for use, handling and storage”).

The IAEA is monitoring the long-term stability of the material and customers will be informed in case of any observed change.

Legal disclaimer

The IAEA shall under no circumstances be liable for any loss, damage, liability, or expense incurred or suffered that is claimed to have resulted from the use of information contained in this Reference Material Certificate.

Citation of this reference sheet

It is suggested to cite this reference sheet according to the following example, as appropriate to the citation format used: INTERNATIONAL ATOMIC ENERGY AGENCY, Reference Sheet, Reference Material IAEA-CH-6, IAEA, Vienna, 5 pp. (The latest version published applies, see “Note” below).

Note

Reference values as stated in this reference sheet may be updated if more information becomes available. Users of this material should ensure that the reference sheet in their possession is current. The current version can be found in the IAEA's Reference Materials online catalogue:

<https://analytical-reference-materials.iaea.org/stable-isotopes> Further information:

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References

- [1] SKRZYPEK, G. et al., Minimum requirements for publishing hydrogen, carbon, nitrogen, oxygen and sulfur stable-isotope delta results (IUPAC Technical Report), *Pure Appl. Chem.* (2022) 1.
- [2] COPLEN, T.B., Guidelines and recommended terms for expression of stable-isotope-ratio and gas-ratio measurement results, *Rapid Commun. Mass Spectrom.* **25** 17 (2011) 2538.
- [3] COPLEN, T.B. et al., New guidelines for $\delta^{13}\text{C}$ measurements, *Anal. Chem.* **78** (2006) 2439.
- [4] HUT, G., Stable Isotope Reference Samples for Geochemical and Hydrological Investigations (Rep. Consultants Group Meeting, Vienna, 1985), Rep. Consultants Group Meeting, International Atomic Energy Agency, Vienna (1987) pp. 1–42.
- [5] ROZANSKI, K., Report on Consultants' Group Meeting on ^{14}C Reference Materials for Radiocarbon Laboratories, IAEA internal report, International Atomic Energy Agency, Vienna (1991) p. 54.
- [6] CHARTRAND, M.M. et al., Final report for CCQM-P212: coherence of carbon isotope delta reference materials, *Metrologia* **60** 1A (2023) 08028.
- [7] BRAND, W.A., COPLEN, T.B., VOGL, J., ROSNER, M., PROHASKA, T., Assessment of international reference materials for isotope-ratio analysis (IUPAC Technical Report), *Pure Appl. Chem.* **86** (2014) 425.
- [8] F. CAMIN, D. BESIC, P.J. BREWER, C.E. ALLISON, T.B. COPLEN, P.J.H. DUNN, M. GEHRE, M. GRÖNINGER, H.A.J. MEIJER, J.F. HÉLIE, P. IACUMIN, R. KRAFT, B. KRAJNC, S. KÜMMEL, S. LEE, J. MEIJA, Z. MESTER, J. MOHN, H. MOOSSEN, H. QI, G. SKRZYPEK, P. SPERLICH, J. VIALON, L.I. WASSENAAR, R.I. WIELGOSZ, Stable Isotope Reference Materials and Scale Definitions – outcomes of the 2024 IAEA experts meeting, RCM, under review.

Revision information

Rev. 01 (2024-12-12) of Rev. 00 of NBS 22, IAEA-CH-3, IAEA-CH-6, IAEA-CH-7, USGS24 reference sheet issue date: 3 August 2007. Changes made: Split into individual reference sheets per material, here for IAEA-CH-6, revision of the assigned value and uncertainty and

extension of validity. The major change in the revision is the correction of the uncertainty according to the characterisation study [3], inclusion of homogeneity and stability sections and of expert`s decision about carbon isotope delta scale.

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The project officer is responsible for the content of this reference sheet.

The Chair of the RM Certification Committee approves this reference sheet and authorizes its release on behalf of the Committee.